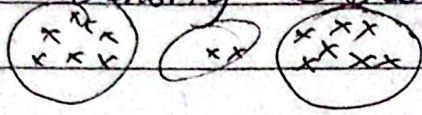
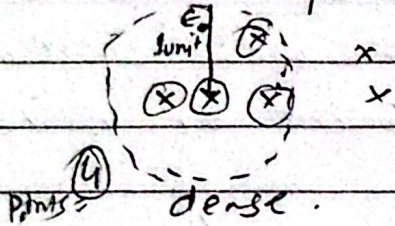


DBSCAN: [Density Based Spatial Clustering of Applications with Noise]

What is Density Based Clustering.



Dense or Sparse



$\epsilon = 1 \text{ unit}$

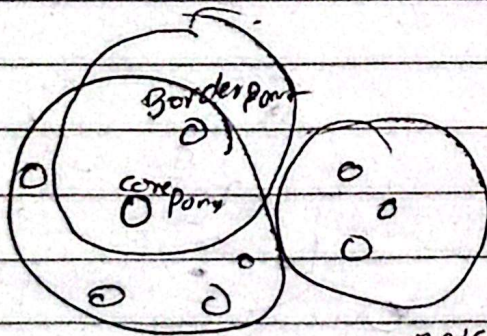
$\text{minpts} \rightarrow 3$

$\rightarrow$ dense

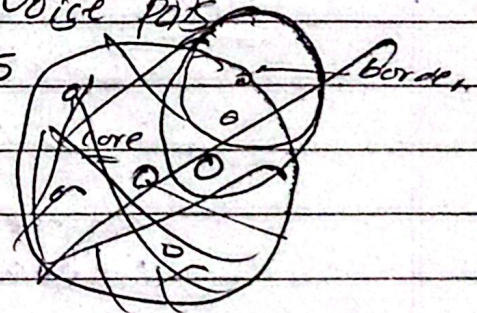
otherwise, sparse.

Core Points, Border Points & Noise points

must be minimum no. of points = 5
to be core point



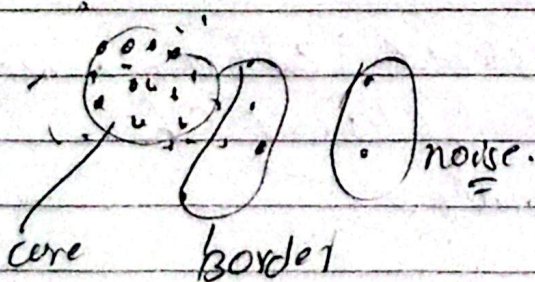
noise point.



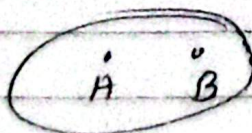
Border Point.

must contain the core in its circle.

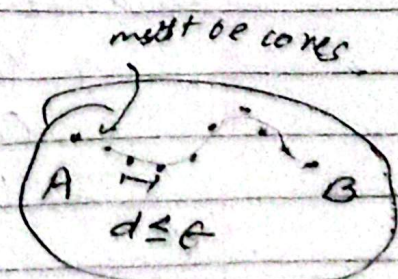
noise point $\rightarrow$ neither core point nor border.



Density Connected Points.



in same cluster.



DBSCAN Algorithm:

Step 0: value define $\rightarrow$ minpts & epsilon.

Let 3

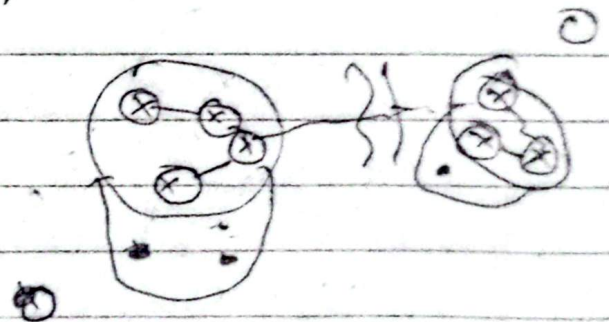
Step 1: Identify all points as either core point, border point or noise point.

Step 2: For all the undustered core points.

Step 2a: Create a new cluster

Step 2b: add all the points that are unclustered and density connected to the current point into the cluster.

x - core
 • - border
 o - noise



Step 3: For each unclustered border point assign it to the cluster of nearest core points.

Advantages

- Robust to outliers
- Only 2 hyperparameters.

Disadvantage - doesn't predict.

- difficult with varying density clusters.